

Hyeonji Bae

Graduate Researcher, E-BRL (Ewha BioRobotics Lab)

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Research Interests: Robot control · teleoperation systems · frequency-domain system evaluation · learning-based control with dynamics models

EDUCATION

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| Mar 2025 – Feb 2027 | M.S. (in progress) , Mechanical and Biomedical Engineering
<i>Ewha Womans University, Seoul, South Korea</i> <ul style="list-style-type: none">• Advisor: Prof. Seok Chang Ryu• GPA: 4.26 / 4.3 (to date)• Thesis (in progress): Frequency-response (Bode plot) evaluation of the responsiveness and consistency of a unilateral (no force feedback) teleoperation system, and latency reduction |
| Mar 2021 – Feb 2025 | B.S. , Mechanical and Biomedical Engineering
<i>Ewha Womans University, Seoul, South Korea</i> <ul style="list-style-type: none">• GPA: 3.71 / 4.3 |

RESEARCH EXPERIENCE

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| Mar 2025 – Present | Graduate Researcher (M.S. Student)
<i>E-BRL (Ewha BioRobotics Lab), Ewha Womans University, Seoul, South Korea</i> <ul style="list-style-type: none">• Improving this teleoperation system (e.g., reducing time delay) and quantitatively evaluating its responsiveness and consistency (frequency response, bandwidth, time constant) for the M.S. thesis |
| Jul 2023 – Feb 2025 | Undergraduate Research Intern
<i>E-BRL (Ewha BioRobotics Lab), Ewha Womans University, Seoul, South Korea</i> <ul style="list-style-type: none">• Integrated the leader input device, the surgical-instrument follower robot, and PC-based control software into the teleoperation system used in the IEEE Transactions on Haptics paper listed below |

PUBLICATIONS

Serin Cha*, Yoonsue Choi*, **Hyeonji Bae**, Chaewon Hwang, Seok Chang Ryu. “On the Systematic Design Framework for Intuitive Interfaces in Telerobotic Operations: A Case Study in Surgical Robotics.” *IEEE Transactions on Haptics*, 2026. doi.org/10.1109/TOH.2026.3684095 (* Equal contribution)

PROJECTS

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|------|--|
| 2024 | Camera-Based Self-Moving Rescue Air Mat (Capstone Design) <ul style="list-style-type: none">• Real-time object detection (OpenCV, YOLO) and wheel control |
| 2023 | SAR: Lower-Limb Wearable Robot for Fall Prevention
<i>Supported by the Challenge Semester program, Ewha Womans University (Fall 2023)</i> <ul style="list-style-type: none">• IMU data preprocessing, SVM-based gait classification, and Arduino-based control of a high-torque actuator |

TEACHING & MENTORING

- | | |
|---------------------|---|
| Aug 2025 – Jul 2026 | Research Mentor
<i>R&E Program, Seoul Science High School, Seoul, South Korea</i> <p>Supported a student R&E research project and its subsequent paper revisions, under the primary supervision of the faculty advisor.</p> |
| Fall 2024 | Teaching Assistant, BioRobotics
<i>Ewha Womans University</i> <p>Led about 10 hands-on ROS 1 lab sessions and graded assignments and lab reports.</p> |

HONORS AND AWARDS

2025	Outstanding Ewha Scientist Scholarship (Half Tuition) <i>Ewha Womans University</i> Mar 2025 - Feb 2027 (two years)
2024	Silver Award, 2024 Ewha Engineering Capstone Design Contest (Single-Major Division) <i>Center for Innovation in Engineering Education, College of Engineering, Ewha Womans University</i> Team of five, for the capstone design project on the rescue air mat.

SKILLS & LANGUAGES

Programming	Python, MATLAB
Robotics Middleware	ROS 1
CAD	SolidWorks, Fusion 360
Languages	Korean (Native speaker); English (Working proficiency (TOEFL iBT 97, 2025))
Certificates	TOEFL iBT (Score 97), ETS (2025); Modern Robotics (Northwestern University), Coursera (2023)

REFERENCES

Prof. Seok Chang Ryu, Department of Mechanical and Biomedical Engineering, Ewha Womans University.